SESSION 4 REFERENCE GUIDE: PANDAS BASICS & TABULAR DATA DISCOVERY

This guide provides a reference for loading, inspecting, auditing metadata, and calculating summary statistics on Pandas DataFrames introduced in Session 4.

1. Creating & Loading Pandas DataFrames

Pandas handles tabular data using DataFrames.

```python
import pandas as pd

# Creating a DataFrame from a Python Dictionary
data = {
    "employee_id": [101, 102, 103],
    "department": ["HR", "Sales", "IT"],
    "salary": [75000.0, 82000.0, 91000.0]
}
df = pd.DataFrame(data)

# Loading data from an external CSV file
# df = pd.read_csv("filepath.csv")
```

2. Structural Inspection Methods

Inspect DataFrame geometry before performing calculations.

Method / Attribute	Purpose	Example
`df.head(n)`	View first n rows	`df.head(5)`
`df.tail(n)`	View last n rows	`df.tail(3)`
`df.shape`	Returns (rows, columns) tuple	`df.shape`
`df.columns`	Returns list of column names	`df.columns`

3. Metadata Auditing (`df.info()`)

`df.info()` displays column data types, non-null counts, and memory footprint.

```python
df.info()
```

```
# Common Pandas Data Types (dtypes):
# - int64: Integers (whole numbers)
# - float64: Floating-point numbers (decimals)
# - object: Text strings or mixed types
# - bool: True/False Booleans
# - datetime64: Dates and timestamps
```

4. Summary Statistics (`df.describe()`)

Generates descriptive statistics for numerical columns.

```python
# Summary statistics for numerical columns
df.describe()

# Summary statistics for categorical/text columns
df.describe(include='object')
```

5. Column Selection & Basic Column Aggregations

```python
# Selecting a single column (returns a Series)
salaries = df["salary"]

# Selecting multiple columns (returns a DataFrame)
sub_df = df[["employee_id", "salary"]]

# Basic Series Aggregations
total_sum = df["salary"].sum()
average_val = df["salary"].mean()
max_val = df["salary"].max()
min_val = df["salary"].min()
```

6. The Dataset Discovery Workflow

```
[Load CSV / Data] --> [Check df.shape] --> [Inspect df.head()] --> [Audit df.info()] --> [Summarize df.describe()]
```